

An exact residue criterion for completely alternating rational sequences

Candidate full solution to Question 1.2 of arXiv:2406.13291

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Status. Candidate full solution, likely valid. The source's general partial-sum conditions are upgraded to a necessary-and-sufficient condition.

Source question

Bhattacharjee and Nailwal [1] ask which sequences $\{r_k(n)\}_{n \geq 0}$ are completely alternating when

$$r_k(x) = \frac{p(x)}{\prod_{i=1}^k (x + b_i)}, \quad 0 < b_1 < \dots < b_k, \quad \deg p \leq k + 1.$$

to this fact, in this article, we work with completely alternating functions which are positive.

Recall that the measure in (1.1) is called the *representing measure* of ϕ and it is unique. This is a consequence of the n -dimensional Weierstrass theorem and the Riesz representation theorem. In [14], F. Hausdorff characterized a completely monotone sequence by Hausdorff moment sequence and the above characterization about completely alternating sequence was first obtained as a special case of the Levy-Khinchin representation theory on abelian semi-groups (see [10], [16]). For a comprehensive study on these topics, we refer the reader to [10], [12], [16], [17].

A lot of work has been done on the characterization of completely monotone functions in terms of Hausdorff moment sequences, for instance, [11, 13, 15] and references therein. However, the characterization of a completely monotone sequence induced by rational functions in terms of zeroes and poles is challenging and seems to be beyond the limit of present understanding. One of the important results in this direction is the sufficient conditions to identify completely monotone sequences introduced by Ball [8] which are solely dependent on the zeroes and poles of the rational functions (see Theorem 1.6). This article aims to determine certain *completely alternating* sequences interpolated by rational functions with real zeros and poles in terms of sufficient conditions. In this article, we deal with the following class of rational functions, that is, for $k \in \mathbb{N}$

$$(1.3) \quad r_k(x) := \frac{p(x)}{\prod_{j=1}^k (x + b_j)},$$

where p is a polynomial of degree at most $k + 1$ and $0 < b_1 < b_2 < \dots < b_k$.

Here, more precisely, we concentrate on the following moment problem:

Question 1.2. *Among all the sequences $\{r_k(n)\}_{n \in \mathbb{Z}_+}$ of rational functions $r_k(x)$, defined in (1.3), which are completely alternating?*

From the operator theoretic aspect, completely monotone and completely alternating sequences are used in revealing the connection with subnormal and completely hyperexpansive operators, respectively (see [1], [2], [3], [4], [7], [6], [9]).

The screenshot is Question 1.2 on page 2 of the source PDF. The source proves a sufficient condition in terms of partial sums of the residues, but not a general necessary-and-sufficient statement.

Proof intuition

Take one forward difference. Complete alternation of $r(n)$ is exactly complete monotonicity of $d_n = r(n+1) - r(n)$. Partial fractions express d_n as the moments of one explicit finite signed measure on $[0, 1]$. The Hausdorff moment theorem says that d is completely monotone precisely when it has a positive representing measure, and polynomial density makes that measure unique even among finite signed measures. Positivity can therefore be read off directly: the linear term supplies a nonnegative atom at 1, and the residue exponential polynomial must have the opposite sign on $(0, 1)$.

Exact classification

Let $p \in \mathbb{R}[x]$ and put $q(x) = \prod_{i=1}^k (x + b_i)$. The partial-fraction decomposition has the unique form

$$\frac{p(x)}{q(x)} = \beta + \alpha x + \sum_{i=1}^k \frac{c_i}{x + b_i}, \quad c_i = \frac{p(-b_i)}{\prod_{j \neq i} (b_j - b_i)}. \quad (1)$$

Here $\alpha = 0$ if $\deg p \leq k$, while for $\deg p = k + 1$ it is the leading coefficient of p . Define

$$w(t) = \sum_{i=1}^k c_i t^{b_i} \quad (0 < t < 1).$$

Use the source convention $\nabla = I - E$, where $(Ea)_n = a_{n+1}$. Thus a sequence a is completely alternating when $\nabla^m a_n \leq 0$ for all $m \geq 1, n \geq 0$.

Theorem 1 (complete answer to Question 1.2). *For $r = p/q$ as above, the sequence $\{r(n)\}_{n \geq 0}$ is completely alternating if and only if*

$$\alpha \geq 0 \quad \text{and} \quad w(t) \leq 0 \quad \text{for every } 0 < t < 1. \quad (2)$$

If “completely alternating” is additionally required to mean strictly positive, as in the source’s working convention, one adds the single condition $r(0) > 0$. For nonnegativity, replace it by $r(0) \geq 0$.

Proof. Set $d_n = r(n+1) - r(n) = -\nabla r(n)$. For every $m \geq 0$,

$$\nabla^m d_n = -\nabla^{m+1} r(n).$$

Consequently $r(n)$ is completely alternating if and only if $d = (d_n)$ is completely monotone.

For $b > 0$,

$$\frac{1}{n+b} = \int_0^1 t^{n+b-1} dt.$$

Equation (1) therefore gives

$$r(n) = \beta + \alpha n + \int_0^1 t^n \frac{w(t)}{t} dt.$$

The integral is finite because every $b_i > 0$. Taking the forward difference,

$$\begin{aligned} d_n &= \alpha + \int_0^1 t^n (t-1) \frac{w(t)}{t} dt \\ &= \int_{[0,1]} t^n d\sigma(t), \end{aligned}$$

where

$$d\sigma = \alpha d\delta_1 - \frac{(1-t)w(t)}{t} dt. \quad (3)$$

This is a finite signed measure.

If (2) holds, (3) is positive. Its moments d_n form a completely monotone sequence by the Hausdorff moment theorem, so $r(n)$ is completely alternating.

Conversely, suppose $r(n)$ is completely alternating. Then d is completely monotone, so the Hausdorff theorem supplies a finite positive Borel measure μ on $[0, 1]$ with $d_n = \int t^n d\mu(t)$ for every $n \geq 0$. The finite signed measures μ and σ have identical moments. Hence they agree on all polynomials, and by Weierstrass approximation they agree on every continuous function. Thus $\mu = \sigma$. Positivity of the atom at 1 gives $\alpha \geq 0$, and positivity of the

absolutely continuous part gives $w(t) \leq 0$ almost everywhere on $(0, 1)$. Since w is continuous there, the inequality holds everywhere.

Finally, the first-order condition makes $r(n)$ increasing. Hence all terms are strictly positive (respectively nonnegative) exactly when the first one is. $\square$

Relation to the source condition and an effective test

Let $C_\ell = \sum_{i=1}^\ell c_i$. Abel summation gives

$$w(t) = \sum_{\ell=1}^{k-1} C_\ell (t^{b_\ell} - t^{b_{\ell+1}}) + C_k t^{b_k}.$$

Thus the source hypotheses $C_\ell \leq 0$ imply $w(t) \leq 0$, recovering its general sufficient theorem. They are not necessary.

Example 2 (strict extension of the source test). Take $b = (1, 2, 3)$ and $c = (-1, 2, -1)$. Then

$$w(t) = -t + 2t^2 - t^3 = -t(1-t)^2 \leq 0,$$

although $C_2 = 1 > 0$. These residues arise from

$$r(x) = 1 - \frac{2}{(x+1)(x+2)(x+3)}.$$

Since $r(0) = 2/3 > 0$, Theorem 1 gives a positive completely alternating sequence that is not detected by the source partial-sum criterion. Indeed,

$$r(n+1) - r(n) = \int_0^1 t^n (1-t)^3 dt.$$

Corollary 3 (exact finite procedure for rational data). If $b_i \in \mathbb{Q}_{>0}$ and p has rational coefficients, the criterion in Theorem 1 is decidable by exact rational arithmetic and a Sturm sequence.

Proof. Choose $L \in \mathbb{N}$ so that $m_i = Lb_i \in \mathbb{N}$. Substituting $t = u^L$ gives

$$w(t) = P(u) := \sum_i c_i u^{m_i}.$$

All c_i are rational. Hence $w \leq 0$ on $(0, 1)$ exactly when the rational polynomial P is nonpositive on $(0, 1)$, which is decidable by isolating its real roots (or the roots of P') with Sturm's theorem and checking the finitely many resulting intervals. $\square$

Verification, scope, and novelty

The proof separately checks the atom at 1, integrability at 0, and the source's positivity convention. Example 2 verifies that the criterion is a genuine strengthening rather than a restatement of the source partial-sum hypothesis. No numerical computation is used.

The answer is exact in terms of the residues and poles. For arbitrary real exponents it leaves a sign test for a finite exponential polynomial; this is still a necessary-and-sufficient classification, and for rational data the corollary makes it a finite exact algorithm.

A bounded novelty search on 11 August 2026 covered the run indexes and exact-phrase/residue-keyword web searches. It found the source and published version, which state only sufficient general conditions, but no later exact answer. Novelty confidence is moderate: the mechanism is a short direct completion of the source's signed-measure calculation and may be implicit folklore, but the theorem was not located or stated in the source.

Human-review recommendation. Verify the signed Hausdorff-measure uniqueness step and confirm that Question 1.2 accepts an exact exponential-polynomial sign criterion rather than requiring a finite list of ordered zero/pole inequalities.

References

- [1] M. Bhattacharjee and R. Nailwal, *A characterization of completely alternating functions*, J. Math. Anal. Appl. (2025); arXiv:2406.13291.
- [2] C. Berg, J. P. R. Christensen, and P. Ressel, *Harmonic Analysis on Semigroups: Theory of Positive Definite and Related Functions*, Graduate Texts in Mathematics 100, Springer, 1984.